

1. The Lagrange method can be extended to problems with two side conditions. If x_0 gives rise to a maximum (or minimum) of $f(x)$ subject to the two side conditions $g(x) = 0$ and $h(x) = 0$, and if $\nabla g(x_0)$, $\nabla h(x_0)$ are nonzero and nonparallel, then there exist scalars λ and μ such that

$$\nabla f(x_0) = \lambda \nabla g(x_0) + \mu \nabla h(x_0).$$

Assume this result and solve the following problems.

- (a) Find the extreme values of $f(x, y, z) = xy + z^2$ subject to the side conditions: $x^2 + y^2 + z^2 = 4$ and $y - x = 0$.
- (b) The planes $x + 2y + 3z = 0$ and $2x + 3y + z = 4$ intersect in a straight line. Find the point on that line that is closest to the origin.
2. Closed rectangular boxes 16 cubic feet in volume are to be constructed from three types of metal. The cost of the metal for the bottom of the box is \$0.50 per square foot, for the sides of the box \$0.25 per square foot, for the top \$0.10 per square foot. Find the dimensions that minimize cost of material.
3. Find the directional derivative of $f(x, y) = e^{2x}(\cos y - \sin y)$ at the point $(\pi/2, -\pi/2)$ in the direction in which f increases most rapidly..
4. Determine the path of steepest descent along the surface $z = 4x^2 + y^2$ from each of the following points: (a) $(1, 1, 5)$; (b) $(1, -2, 8)$.